Code No: 157ER **R18**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech IV Year I Semester Examinations, January/February - 2023
HEALTH AND SAFETY IN MINES
(Common to EEE, CSE)

Time: 3 Hours **Max.Marks:75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

PART – A

(25 Marks)

1.a)	What accidents come under the preview of the Mine Act?	[2]
b)	List the type of accidents in mines prone.	[3]
c)	Present briefly the cost of accidents.	[2]
d)	Discuss the overview of opencast accidents in Indian coal mines.	[3]
e)	List the techniques of safety analysis.	[2]
f)	Explain with examples, emergency contingency plan.	[3]
g)	What is Safety management?	[2]
h)	State the objective of Auditing.	[3]
i)	What is the role of the Safety Professional in mines?	[2]
j)	Briefly present the scope and responsibilities of safety polices.	[3]

PART – B

(50 Marks)

2.a) Explain the reason to preventing accidents in mines, discuss in detailed examples.
b) What is the importance of statistical analysis of mine accidents? [5+5]

OR

3.a) Illustrate the role and responsibilities of safety and health professionals.
b) What are the causes of accidents in underground coalmines due to fall of roof, and how to prevent them? [5+5]

4.a) Classify an accident enquiry report for accident due to blasting in open cast mines.
b) Depict the various causes and preventive measures of accidents due to fall of roof and sides. [5+5]

OR

5.a) Enumerate the statistical analysis of accidents and their importance for promotion of safety, classify it.
b) Illustrate causes and preventive measures of accidents due to movement of machinery in opencast mines. [5+5]

6.a) Analyze the System engineering approach to safety Techniques used in safety analysis.
b) Depict and analyze the generic approach to loss control within mining operations.[5+5]

OR

7.a) Write short note on the concept of ZAP and MAP.
b) What are the safety techniques used in mines? [5+5]

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8.a) Discuss in detailed process of occupational diseases in mining.
b) What is FTA, Explain about Fault tree analysis? [5+5]

OR

9.a) Explain about the various risk elimination methods used in analysis.
b) Enumerate the Risk estimation and evaluation methods used in mines. [5+5]

10.a) Explain the implementation safety policies and Describe what factors to consider for disaster management in mines?
b) What is internal safety organization and discuss various development of safety consciousness in mines. [5+5]

OR

11.a) Discuss the implementation of Safety Management System in Indian mines.
b) Illustrate the occupational Safety and Health Act and enumerate Origin of OSHA Standards. [5+5]

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